

LESSON 7-2 • SESSION 2 • PRACTICE • FORM B

Opposites of Rational Numbers

Learning target – I can find the opposite of a rational number and explain what taking the opposite twice does. • 6.NS.C.6a

REMEMBER

- **Opposites** are the same distance from 0 on different sides: 2.5 and -2.5 . Zero is its own opposite.
- $-(4.5)$ means **the opposite of 4.5**, so $-(4.5) = -4.5$ and $-(-4.5) = 4.5$.
- Taking the opposite **twice** gives back the number you started with.

WORKED EXAMPLEFind $-(-5.3)$. Then find the opposite of your answer.**Step 1** -5.3 is 5.3 units left of 0.**Step 2** Flip to the other side of 0, keeping the distance 5.3: $-(-5.3) = 5.3$.**Step 3** Flip again: the opposite of 5.3 is -5.3 .**Answer** 5.3, then -5.3 (two flips return to the start)**WORDS** **opposite** (*opuesto*) • **opposite of the opposite** (*opuesto del opuesto*) • **rational number** (*número racional*)**PRACTICE**

show your work • label units

1. Write the opposite of each number, or find the value.

a. -3.5 _____

b. $3/10$ _____

c. $-3/4$ _____

d. 0 _____

e. $-(6.2) =$ _____

f. $-(-6.2) =$ _____

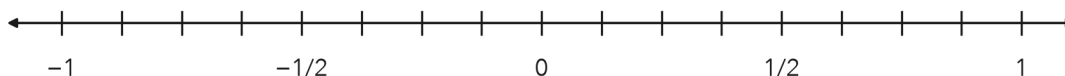
- 2.
- Scuba Dive Log.**
- A dive log lists four depths: reef
- -3.5
- m, ledge
- -6.25
- m, wreck
- -10.5
- m, sandbar
- -1.2
- m.

a. Write the opposite of each depth. _____

b. How far from 0 is the ledge depth? _____ m

Show your work.

- 3.
- Card Flips.**
- Omar wrote
- $-5/8$
- on a card. He flipped it to its opposite, then flipped that result to its opposite. What number does he have now?

Hint Do one flip at a time and mark where each lands.**Answer** _____

- 4.
- Club Balance.**
- A club balance dropped to
- -24.3
- dollars in March. In April it changed to the opposite amount. What is the April balance?

Hint Same distance from 0, other side.

Show your work.

Answer _____ dollars

- 5.
- Missing Start.**
- A number's opposite is
- -3.6
- . What number did you start with?

Hint Flip back: take the opposite of -3.6 .

Show your work.

Start _____

- 6. Ladder and Pit.** A ladder top reaches 5.25 feet above a shop floor (0). A pit bottom sits at the opposite value.
- a.** Write the number for the pit bottom. _____ ft
- b.** How far is each one from the floor? _____ ft

Show your work.

- 7. Twice Over.** What number is the opposite of the opposite of -8.15 ?

Hint Two flips land where you started.

Show your work.

Answer _____

- 8. Three Readings.** Three sensors read -3.5 , 3.5 and -0.25 degrees.

- a.** Write the opposite of each reading. _____
- b.** Which original reading is closest to 0? _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Diego's Work.** Diego wrote: " $-(-7.4) = -7.4$, because two negatives make the number more negative." Explain his mistake and give the correct value.

TEST PRACTICE

choose the best answer

- 10.** $-(-2.6)$ equals –
- A** -2.6 **B** 2.6 **C** 0 **D** -0.26
- 11.** The opposite of $-3/7$ is –
- A** $-3/7$ **B** $7/3$ **C** $3/7$ **D** $-7/3$

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

$-5/8$ L	6.25 F	-8.15 E	-3.6 X	$5/8$ Q	-5.25 P	-0.25 D	24.3 I	3.6 P	-24.3 Z
#2b		#3	#4	#5	#6a	#7	#8b		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____